

Birzeit University

Second Semester, 2025/2026

Assignment 1: Titanic Survival Classification Using Machine Learning

Due Date May 30, 2026

In this case study, you will build and evaluate machine learning classifiers for predicting passenger survival on the Titanic dataset. The focus is on applying a clear preprocessing pipeline, training classification models, tuning their parameters, and comparing their performance.

Code to load dataset:

```
import seaborn as sns
import numpy as np

# load dataset titanic
df = sns.load_dataset('titanic')
```

Tasks

1. Load and Explore the Dataset

- Load the Titanic dataset using the provided code snippet.
- Perform a brief exploratory analysis:
 - Display dataset structure and feature types.
 - Identify missing values.
 - Show basic statistics and class distribution.

2. Data Preprocessing

- Handle missing values using suitable techniques (e.g., mean/median imputation or mode filling).
- Encode categorical features using methods such as one-hot encoding or label encoding when needed.
- Normalize or standardize numerical features when needed.

3. Feature Selection

- Apply a simple feature selection technique to identify important features, such as:
 - Correlation analysis,
 - Information gain, ...

4. Train/Test Split

- Split the dataset into training, validation, and testing sets for evaluation.

5. Classification Models

- Train the following classifiers:
 - Support Vector Machine (SVM)
 - Random Forest

6. Hyperparameter Tuning

- Improve model performance using parameter tuning techniques such as Grid Search.
- Example parameters:
 - **SVM:** kernel, C, gamma
 - **Random Forest:** number of trees, max depth, min samples split

7. Model Evaluation and Comparison

- Evaluate both models using suitable metrics such as:
 - Accuracy
 - Precision
 - Recall
 - F1-score
- Perform the following comparisons:
 - SVM vs. Random Forest
 - Default parameters vs. tuned parameters
 - Performance before preprocessing and after applying the preprocessing pipeline
- Briefly discuss:
 - Which model achieved the best performance
 - The impact of preprocessing on classification results
 - The effect of parameter tuning on model performance

Recommended Tools and Libraries

You may use Python libraries such as:

- **Pandas** for data loading and preprocessing
- **NumPy** for numerical operations
- **Matplotlib** or **Seaborn** for visualization
- **Scikit-learn (sklearn)** for:
 - preprocessing,
 - feature selection,
 - train/test splitting,
 - SVM and Random Forest classifiers,
 - hyperparameter tuning,
 - evaluation metrics

Submissions

- Submit your work via Ritaj as a reply to this message
- Submit the solution python code.
- Submit a short report describing:
 - Preprocessing steps,
 - Feature selection method,
 - Parameter tuning process,
 - Evaluation results and comparison.